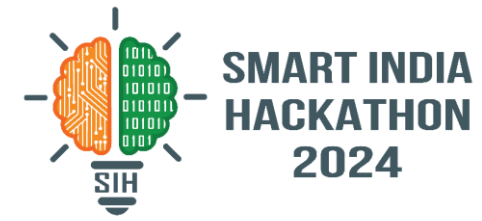
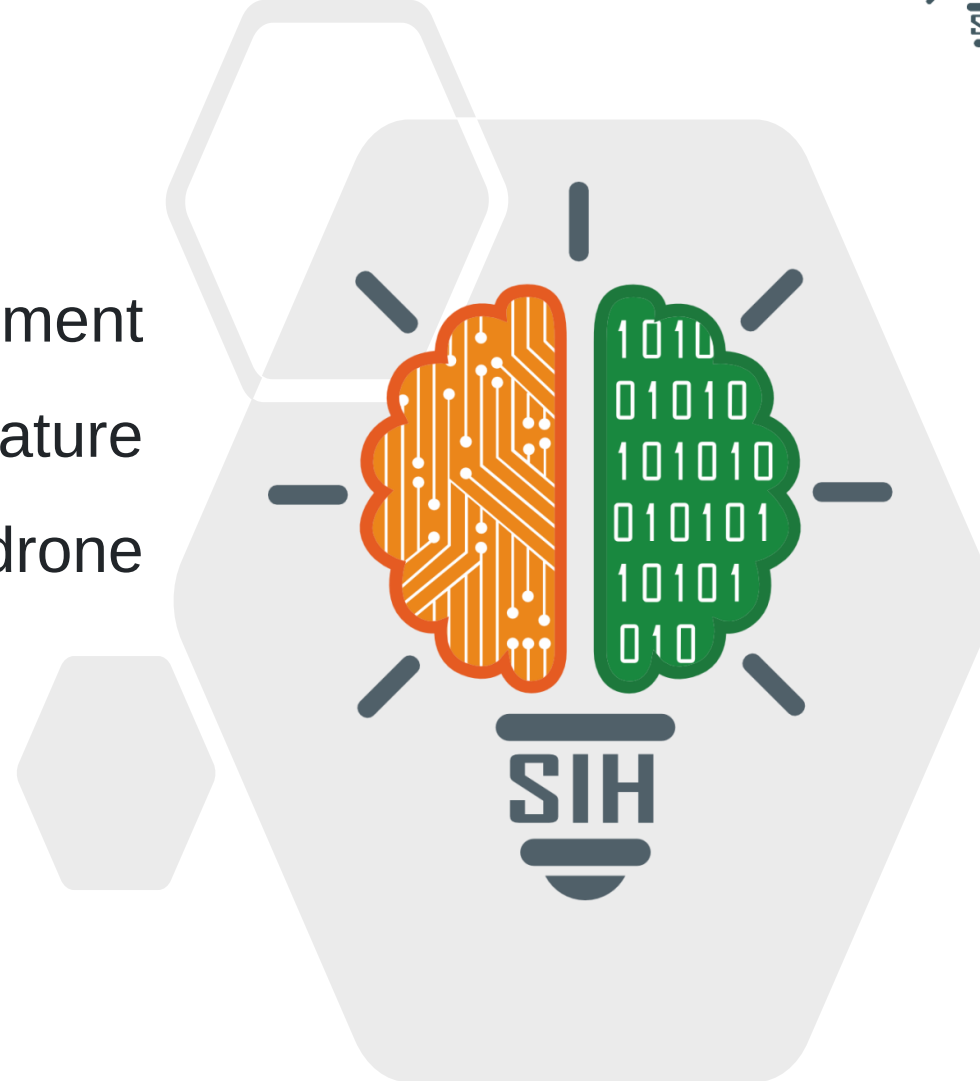


# SMART INDIA HACKATHON 2024



- **Problem Statement ID** - 1705
- **Problem Statement Title** - Development and Optimization of AI model for Feature identification/ Extraction from drone orthophotos.
- **Theme** - Robotics and Drones
- **PS Category** - Software
- **Team ID** - WCOE20
- **Team Name** - Team CyberPunk



# AI Model for Feature Extraction from Drone Imagery



## ❖ Proposed Solution : -

- **Detailed Explanation :**

- This is a classic problem of semantic segmentation of aerial imagery data.
- Develop an CNN model to automatically identify key features such as building footprints, road features, and waterbodies from SVAMITVA drone imagery dataset .
- The model used is derived from **U-Net model** architecture.

- **How it Addresses the Problem :**

- The solution will eliminate manual land surveying, reducing human errors and time consumption. This will create precise digital records and enhance the efficiency of rural land management.

- **Innovation and Uniqueness :**

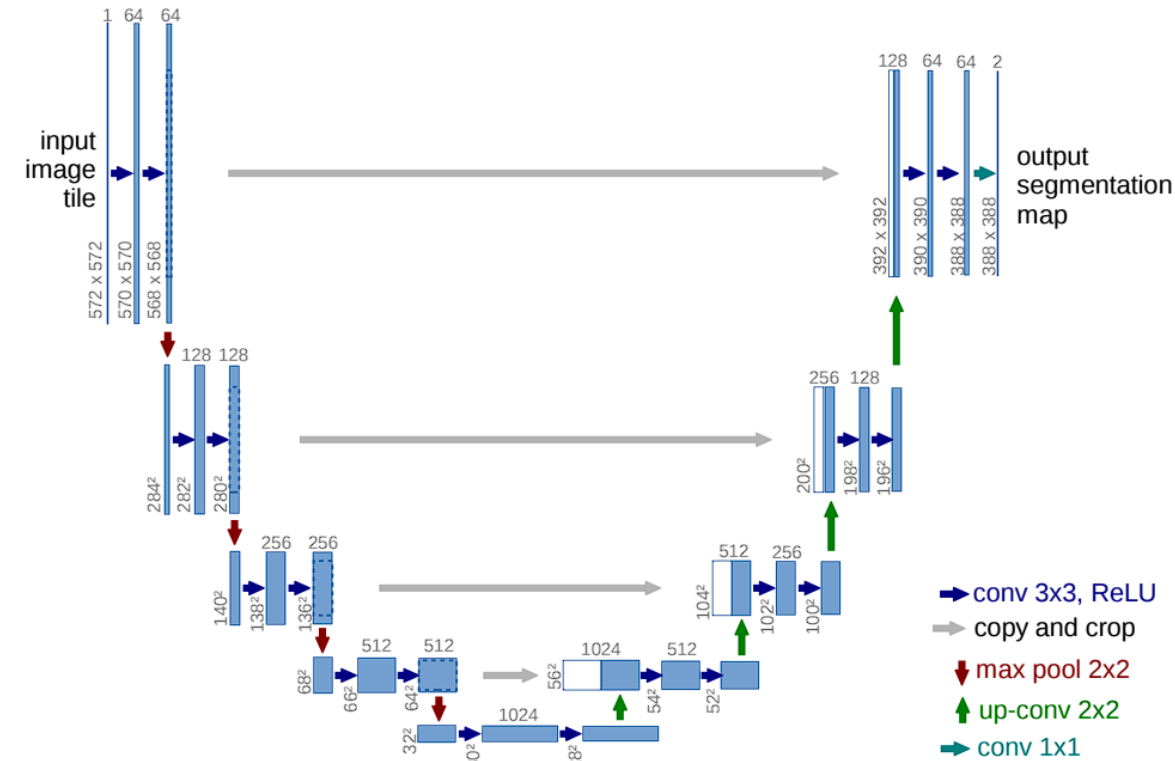
- The model will operate on a cloud-based platform, enabling wide-scale accessibility and deployment. Achieving high accuracy (95%) ensures reliable feature extraction for rural planning purposes.

## Methodology

- Utilize the SVAMITVA Scheme drone-labeled datasets for 10 villages to train and validate the model.
- The U-net model widely used for image segmentation problems is employed for this project and transfer learning is used.
- Train the model for extracting various features (roads, water bodies, roofs, etc.) and then fine tuning it for the provided dataset.
- Test the model on other drone orthomosaics to test its performance.

## Technologies Used:

- Programming Languages: Python
- Frameworks & Libraries :TensorFlow, Keras, Django, GDAL, Geopandas
- Tools: Google Colab, Jupyter Notebooks, Kaggle
- Frontend : HTML5, CSS, JS



- **Feasibility**

- Access to accurate and high-resolution SVAMITVA drone data, which provides clear and labeled images.
- Existing technology, frameworks and cloud infrastructure, supports model development and deployment.

- **Challenges**

- Achieving a target accuracy of 95% while maintaining efficient processing speed requires careful tuning of the model.
- Model size should be as small as possible for implementing on edge devices or browsers.

- **Overcoming Strategies**

- We will optimize the model by reducing complexity fine-tuning and tuning hyperparameters.
- Usage of regularization techniques (dropout layers, L1 & L2 regularization) and model pruning.

# IMPACT AND BENEFITS

- **Impact :**
  - This supports better rural planning, such as solar installations and property tax assessments, and aids in resolving property disputes, thereby benefiting the community.
- **Benefits :**
  - By improving the accuracy of land records, the solution will help property owners access bank loans and enable village-level development projects.
  - It will also enhance the quality of life in rural areas by supporting sustainable development and efficient resource management.





- AI-Assisted Manual Segmentation Web Application for Geospatial Satellite and Imagery Data - Lingyan Rana, , Cheng Jib, Shizhou Zhangb,c, Xiaoqiang Zhangd , Yanning Zhang
- An Unsupervised Domain Adaption Framework for Aerial Image Semantic Segmentation Based on Curriculum Learning - Xun Thong Cham, Chen Fung Choy Oliver
- A brief introduction to satellite image segmentation with neural networks — Robin Cole ( Medium)